

The sound of silence: Does competition decrease willingness to blow the whistle? Evidence from a preregistered experiment*

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Abstract

We investigate whether market competition suppresses employees' willingness to report managerial misconduct, using a preregistered laboratory experiment with two participant pools ($N = 240$). In a setting where firms either operate independently or compete in a tournament for market revenue, managers choose between a productive task and breaking the law to double firm surplus at the expense of the public, while employees decide whether they would report the manager if the manager chose to break the law. At the aggregate level, competition does not significantly reduce employees' conditional willingness to blow the whistle. Although manager lawbreaking increases directionally under competition, this is not accompanied by a statistically significant decline in employees' stated willingness to report misconduct. Exploratory analyses suggest possible heterogeneity across participant pools for employee reporting but not for manager lawbreaking.

Keywords: whistleblowing, market competition, corporate fraud, moral erosion, experiment

JEL Codes: C92, D63, D91, G33, K42, M14

1 Introduction

Market competition is the central organising principle of modern economies, and is traditionally celebrated for driving efficiency and innovation. Classical economic theory posits that competition forces firms to optimise, driving out inefficient practices and fostering trust through repeated mutually beneficial trade. At the same time, a long-standing counterargument suggests that competition may weaken ethical constraints. Shleifer (2004) famously argued that in highly competitive markets, firms may be pressured to adopt unethical cost-cutting practices, such as fraud and corruption, simply to survive. In this view, unethical behaviour need not reflect only individual greed, but may arise as a structural consequence of a race to the bottom.

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Whether this logic applies empirically, however, is far from obvious. On the one hand, several experimental studies find that market interaction can erode moral concerns relative to individual decision-making (Falk and Szech, 2013; Bartling et al., 2015; Ziegler et al., 2024). On the other hand, moral behaviour is often more robust than simple material incentives would predict. In experimental cheating tasks, many participants do not fully exploit private opportunities to lie for monetary gain (Fischbacher and Föllmi-Heusi, 2013), and meta-analytic evidence across truth-telling experiments suggests that people lie surprisingly little, consistent with preferences for honesty and for appearing honest (Abeler et al., 2019). In the whistleblowing domain, Choo et al. (2019) show that whistleblowing depends on pivotality, diffusion of responsibility, and whether the motive for reporting is selfish or altruistic. Thus, moral behaviour under competitive or strategic pressure may depend strongly on whether individuals feel pivotal, whether reporting is self-serving or prosocial, and whether responsibility can be shifted to others. Recent evidence from a large-scale crowdsourced experiment similarly suggests that competition has at most a small adverse average effect on moral behaviour, with substantial heterogeneity across designs (Huber et al., 2023). The effect of competition on whistleblowing is therefore theoretically ambiguous.

This ambiguity is particularly critical in the context of corporate fraud. Dyck et al. (2023) estimate that out of all large publicly traded firms, 41% engage in financial accounting frauds and 10% engage in securities fraud, imposing yearly welfare losses of \$830 billion in 2021 in the United States alone. Despite its substantial welfare costs and the erosion of public trust in the financial system, only a third of all corporate fraud cases are ever detected (Dyck et al., 2023). Consequently, whistleblowing, defined as the disclosure of illegitimate practices by organisation members, has emerged as a vital tool to expose and deter fraudulent activities (Berger and Lee, 2022; Leder-Luis, 2023). However, the decision to blow the whistle is rarely a simple cost-benefit consideration of legal incentives; it represents a fundamental dilemma, pitting the moral responsibility to prevent social harm against the powerful impulse of loyalty to the firm (Mesmer-Magnus and Viswesvaran, 2005; Waytz et al., 2013; Dungan et al., 2019).

Examining whistleblowing empirically is inherently complex and faces identification and measurement challenges, as only detected fraud and blown whistles can be observed. To address these issues, researchers have increasingly relied on experiments to provide empirical evidence for emerging whistleblowing laws.¹ The experimental literature on whistleblowing has primarily focused on pecuniary factors, such as monetary rewards, protection from dismissal, and the threat of fines (Breuer, 2013; Butler et al., 2020; Mechtenberg et al., 2020; Schmolke and Utikal, 2025). Beyond these incentives, researchers have explored non-pecuniary motivations, including personality traits, diffusion of responsibility, trust in institutions, anticipation of social approval, and social norms (Bartuli et al., 2016; Choo et al., 2019; Antinyan et al., 2020; Butler et al., 2020; Djawadi et al., 2025). A common limitation in these studies, however, is that they focus on the behaviour of experimental firms in isolation. This abstraction ignores the reality that competitive pressure may undermine whistleblowing and facilitate lawbreaking.

Establishing the relationship between competition and whistleblowing is of significant policy

¹Examples of such laws include the Public Interest Disclosure Act in the United Kingdom (1998), the Dodd-Frank Act in the United States (2010), and the more recent EU Whistleblower Protection Directive in the European Union (2019).

relevance. From the perspective of effective regulation, if competition systematically suppresses whistleblowing, it implies that internal monitoring is most fragile in precisely those sectors where market pressure is most intense. Given that interventions to encourage reporting, such as legal protection from retaliation or financial rewards, are costly to implement², policymakers must prioritise industries where such measures are most effective. Competitive industries are natural candidates for such prioritisation, as competition may increase managers' incentives to break the law and reduce employees' willingness to report misconduct. Thus, establishing whether such pressure exists is vital to maximise the impact of interventions and more effectively protect public welfare from corporate misconduct.

In this paper, we fill this gap by conducting a preregistered experiment to test the robustness of whistleblowing under competitive pressure. We compare whistleblowing and lawbreaking in a setting where firms operate independently to a setting where they compete for market revenue. Our experiment builds upon the whistleblowing game introduced by Butler et al. (2020). In this game, managers choose between a legitimate multiplication task or breaking the law to double firm surplus at the expense of the public. Employees, who benefit from the surplus, state whether they would report their manager if the manager chose to break the law, before observing whether misconduct actually occurred. In addition to these primary behavioural choices, we elicit incentivised beliefs regarding the prevalence of such actions, and social appropriateness judgements.

A key distinction of our setting relative to much of the markets and morals literature is that the treatment changes competitive pressure while holding the whistleblower's pivotality and reporting technology fixed. Employees face the same reporting cost, the same probability of implementation, and the same manager penalty across treatments. The experiment therefore isolates whether competitive pressure itself reduces conditional willingness to report misconduct, rather than combining competition with changes in the likelihood that an individual's report is decisive.

Overall, we find little evidence that competition affects unethical behaviour. Our primary results indicate a statistically null effect of competition on employees' conditional willingness to blow the whistle. Specifically, we observe an insignificant decrease in stated willingness to report managerial lawbreaking under competition, and an insignificant increase in manager lawbreaking. While competition shifted participants' beliefs in the direction of the observed behaviour, it did not significantly affect their underlying social appropriateness judgements. These findings suggest that the act of whistleblowing is relatively robust to competitive pressure, which may imply that market intensity need not be the primary concern for regulators when prioritising enforcement resources.

Though exploratory, we document evidence of heterogeneity across participant pools in conditional willingness to report. The negative effect of competition on whistleblowing is concentrated in the Birmingham pool, whereas the Cambridge pool shows little response. We do not find comparable evidence that the effect of competition on manager lawbreaking differs significantly across pools. These results suggest that aggregate null effects may mask context-sensitive

²For the EU Whistleblower Protection Directive, European Commission (2018) estimates one-off implementation costs of €746.7 million, and recurring annual costs of €1.2 billion, for public and private sectors combined.

responses, although further replication is needed to obtain precise estimates of heterogeneity. Future research could also investigate higher levels of competitive intensity, such as winner-takes-all tournaments where only the top firm retains its surplus, or environments featuring survival pressure where failing firms are eliminated, to test the boundary conditions of the null effect reported here.

The remainder of the paper is organised as follows: Section 2 describes the experimental design and theoretical predictions; Section 3 presents the results, distinguishing between pre-registered analyses and exploratory findings regarding heterogeneity; and Section 4 provides a concluding discussion.

2 Experimental design and predictions

2.1 The baseline whistleblowing game

The game is based on the whistleblowing game introduced by Butler et al. (2020). Each experimental session involves fifteen participants, who are randomly assigned to one of three roles: manager, employee, or member of the public. Nine participants are organised into three firms, each consisting of one manager and two employees, while the remaining six participants represent members of the public.

The game proceeds in three distinct stages. The first stage is the production stage. The two employees engage in a real-effort addition task, where they are asked to report the sum of six pairs of two-digit numbers. For each correctly solved pair, the employee earns 20 ECU as private earnings and contributes 10 ECU to the firm surplus. Employees are given 120 seconds to complete this stage. This structure ensures that the firm’s assets are legitimised by labour, establishing a baseline of moral entitlement before management decisions occur. Members of the public also perform the same task to earn private income, though they do not contribute to any firm surplus.

The second stage is the management stage, which determines the final size of the firm surplus. The manager receives a fixed income of 120 ECU and has the opportunity to double the firm’s surplus via one of two options. The first option is doing a multiplication task. The manager is asked to report the product of six pairs of two-digit numbers. If they succeed in computing at least three products correctly, the firm surplus is doubled; otherwise, it remains unchanged. This option represents lawful compliance, but carries execution risk. The second option is breaking the law. The manager skips the multiplication task, and the firm surplus is doubled automatically. However, this choice generates a negative externality, deducting 20 ECU from the earnings of every member of the public. Importantly, the manager does not know the size of the surplus created by the employees when making this decision, which ensures that the choice to break the law is comparable across firms and independent of employee performance.

The third stage is the whistleblowing stage. Employees do not observe their manager’s choice directly. Instead, using the strategy method, each employee states whether they would blow the whistle if their manager had broken the law. Reporting comes at a private cost of 25 ECU for the employee and imposes a monetary penalty of 70 ECU on the manager. This penalty is deducted from the manager’s individual earnings and does not reduce firm surplus.

To mirror the uncertain nature of regulatory sanctions, one of the two employees is randomly selected, and only their decision is implemented. This decision is a payoff-relevant contingent commitment: if the manager broke the law and the employee is selected for implementation, the stated decision determines whether the employee pays the reporting cost and whether the manager is penalised. If the manager did not break the law, the employee's decision is irrelevant, and no reporting cost is incurred.

Finally, payoffs are determined by the firm's net surplus. The final surplus is distributed among the firm members: the manager receives 50%, and each employee receives 25%. These profit shares are added to the manager's fixed wage and the employees' private task earnings to determine final compensation.

2.2 The competition treatment

The competition treatment introduces a tournament incentive structure that breaks the independence between firms by linking their final payoffs. This treatment is designed to reflect industries where a firm's revenue depends significantly on its performance relative to its rivals. The first three stages of the game proceed exactly as in the baseline. The modification occurs during the determination of final payoffs.

Before redistributing the surpluses to firm members, we rank the post-management firm surpluses generated by the three firms within each session. Whistleblowing penalties are not deducted from firm surplus and therefore do not affect tournament rankings. The firm with the largest post-management surplus is designated the winner, and its surplus is subsequently increased by 50%. The remaining two firms in the session have their surpluses reduced by 25%. In the event of a tie for the largest surplus, the winner is selected randomly with uniform probability.

This structure introduces strategic uncertainty, as firm members must form beliefs about the behaviour and task performance of rival firms. Consequently, competitive pressure provides both motives to decrease whistleblowing and increase lawbreaking, as firm members seek to secure the 50% bonus and avoid the 25% penalty.

2.3 Predictions

We specify two predictions regarding the behaviour of managers and employees across the Baseline and Competition treatments. Formal derivations are provided in Appendix A.

The Competition treatment introduces a significant financial incentive to double the firm's surplus, as doing so increases the firm's probability of winning the tournament and securing the largest payoff share. This increase in the expected material benefit of lawbreaking, relative to the multiplication task, provides managers with a strategic incentive to choose the former. This tendency may be further reinforced if competition reduces the perceived moral costs of breaking the law.

Prediction 1. *Managers' willingness to break the law is higher under competition than in the baseline.*

The comparative static for employees should be interpreted as a prediction for conditional reporting under the strategy-method elicitation used in the experiment, rather than as a prediction about reporting after directly observed misconduct. Under this elicitation, the employee makes a payoff-relevant commitment to report if the manager broke the law. The private cost of this commitment is incurred only if the manager actually broke the law and the employee is selected for implementation. Thus, if competition increases managerial lawbreaking, it also increases the probability that a conditional reporting decision becomes payoff-relevant. This provides a material channel through which competition may reduce conditional willingness to report. Competition may also reduce reporting if it weakens the perceived moral obligation to report, for example by making managerial lawbreaking under competitive pressure seem less culpable or by increasing loyalty to the firm.

Prediction 2. *Employees' conditional willingness to blow the whistle is lower under competition than in the baseline.*

Beyond these primary behavioural outcomes, we consider the impact of competition on the underlying psychological drivers of decision-making, namely beliefs and social appropriateness judgements. We expect the competitive tournament to shift normative expectations, leading participants to believe that lawbreaking will be more prevalent and whistleblowing less frequent among their peers. Furthermore, competition may alter the perceived social appropriateness of these actions, potentially attenuating the moral stigma associated with lawbreaking or staying silent. We anticipate that these internal states will correlate with individual behaviour. Specifically, managers who view lawbreaking as more socially appropriate or more common are expected to be more likely to engage in it, while employees who perceive staying silent as more appropriate or more common should be less inclined to report misconduct.

2.4 Implementation

Each session consisted of exactly 15 participants and was played for 12 rounds. Managers retained their role throughout all rounds, while non-manager participants alternated between employee and member of the public roles. Employees were randomly reassigned to firms in each round, so employee–manager pairings changed over time. Consequently, employee–manager relationships were temporary, limiting repeated-game and reputation effects despite the continuity of managers. Participants always knew their own role, but were never informed of the identities or roles of other participants during the session.

To keep decisions comparable and provide cleaner measures of beliefs, participants did not receive feedback between rounds about managers' lawbreaking decisions or employees' whistleblowing decisions, either in their own firm or in other firms. Feedback was limited to participants' own task performance and, in the Competition treatment, whether their firm won or lost the tournament. Thus, the additional feedback in the Competition treatment concerned only the payoff-relevant competition outcome and did not reveal the behaviour of managers or employees.

Following the game, all participants completed a post-experiment survey. First, we elicit behavioural beliefs, where participants estimated the frequency of whistleblowing and lawbreak-

ing in their session, incentivised for accurate estimates. Next, we elicit social appropriateness judgements regarding a manager breaking the law, an employee not blowing the whistle, and the public losing part of their earnings, as well as participants' loyalty to the firm.³ These judgements were incentivised using the Krupka and Weber (2013) method, rewarding participants for matching the modal response. Finally, we elicit risk preferences using the incentivised Eckel and Grossman (2002) lottery task, and social preferences using the Social Value Orientation scale (Murphy et al., 2011). Standard demographic information was also collected.

The experimental sessions took place between April and October of 2023. Participants were recruited equally from the Birmingham Experimental Economics Laboratory and the Cambridge Experimental and Behavioural Economics Group. The experiment was programmed in oTree (Chen et al., 2016) and preregistered⁴. Ethical approval was obtained prior to data collection.⁵ Instructions were provided describing all roles to ensure common knowledge of the game structure. Participants needed to correctly answer comprehension questions before making decisions.

In total we had 240 participants, 120 per treatment arm. Assuming a baseline whistleblowing probability of 0.23 from Butler et al. (2020) and a Linear Probability Model, the minimum detectable effect size with 120 participants per treatment was 0.10. Within each treatment arm, we had 60 participants from each participant pool. Each participant took part in only one experimental session. Participants received an average payment of £11.21 per session for approximately 75 minutes, including a participation fee of £2.00.

Appendix Table B1 and Table B2 report demographic characteristics, risk preferences, social value orientation, and task performance across treatment-pool cells. Observable characteristics are similar across treatment arms within each participant pool. Addition-task performance is also close to ceiling in all cells. Multiplication-task performance is reported descriptively, but should be interpreted with caution because it is observed only for managers who chose the multiplication task rather than breaking the law.

2.5 Design choices and interpretation

Before turning to the results, it is useful to clarify several design choices that are important for interpreting the treatment comparison. These choices do not affect the internal comparison between Baseline and Competition, but they define the type of whistleblowing environment studied in the experiment.

First, following Butler et al. (2020), whistleblowing in our design does not compensate members of the public for the losses caused by managerial lawbreaking. Instead, it only imposes a monetary penalty on the manager of their firm. This captures settings in which whistleblow-

³The specific norms we measure are motivated by the management and business ethics literature on rationalisations, which are cognitive strategies employed to reconcile one's unethical actions with moral standards (Ashforth and Anand, 2003; Castro et al., 2020). Relevant rationalisations for whistleblowing are: (i) denial of responsibility, where actors justify their actions believing others in similar positions behave the same way, (ii) selective social comparison, in which individuals see others in different positions as more unethical than themselves, (iii) denial of the victim, perceiving the victim as deserving the harm, and (iv) appeal to higher loyalties, such as firm loyalty.

⁴The preregistration is available at: <https://www.socialscienceregistry.org/trials/11051>.

⁵University of Amsterdam ethical approval reference number: EB-953, University of Birmingham ethical approval reference number: ERN-0894-3, University of Cambridge ethical approval reference number: 23-32.

ing triggers punishment or enforcement but does not fully restore victims’ losses, for example because of legal costs, delays, or difficulties identifying victims and measuring individual harm. A possible implication is that baseline reporting may be lower than in environments where whistleblowing directly restores victims’ losses, leaving less scope for competition to further reduce reporting.

Second, the whistleblowing decision is elicited using the strategy method. Employees state whether they would blow the whistle if their manager had broken the law, without observing the manager’s actual choice. This design choice follows closely related experimental work on whistleblowing (Butler et al., 2020), and allows us to observe a reporting decision for every employee in every round, which is important because lawbreaking is relatively infrequent, especially in the baseline treatment. The decision is nevertheless payoff-relevant rather than purely hypothetical: if the manager broke the law and the employee is selected for implementation, the employee’s stated decision determines whether the reporting cost and manager penalty are applied.

This feature is important for interpretation. The elicited outcome is conditional willingness to report misconduct under ambiguity about whether misconduct occurred, not whistleblowing in response to directly observed wrongdoing. The strategy method may attenuate emotional or reactive motives for whistleblowing that arise when misconduct is known with certainty. This concern is especially relevant in punishment decisions, where Brandts and Charness (2011) report that punishment levels tend to be lower under the strategy method than under direct-response elicitation. This may further attenuate reporting levels relative to a direct-response design. Because the elicitation method is held constant across treatments, this feature does not confound the treatment comparison, but it defines the reporting environment to which our estimates apply.

Third, managers retained their role throughout the experiment, while non-manager participants alternated between employee and public roles. This role-assignment structure is similar to Mechtenberg et al. (2020), where employers remain employers throughout the experiment, while other participants switch roles; they explain that this reshuffling was intended to make the negative consequences of misconduct more salient to participants whose decisions may curb the harm imposed by employers. In our setting, the rotation of non-managers serves a similar purpose: participants who make reporting decisions as employees also experience the position of members of the public harmed by lawbreaking in other rounds. Keeping managers fixed preserves a stable managerial decision environment and avoids introducing role-transition effects into the lawbreaking decision. A consequence is that managers do not directly experience the public role, so managerial lawbreaking should be interpreted as behaviour by participants assigned to a stable managerial role.

Finally, as can be seen from the instructions in Appendix C, our study is a framed experiment, where labels for roles and actions matched the description of the game. Alekseev et al. (2017) summarised existing evidence and concluded that “using evocative language either does not affect behaviour or affects it in a desirable way by evoking the desired emotional response”.

3 Results

This section presents both the preregistered analyses and exploratory heterogeneity analyses. Although the theoretical predictions are presented in the order of the underlying causal logic (manager lawbreaking first and employee whistleblowing second) the results section begins with whistleblowing because it is the primary outcome of interest of the paper. Manager lawbreaking is then analysed as an important complementary outcome that helps interpret the whistleblowing results.

3.1 Preregistered analysis

Figure 1 presents mean whistleblowing and lawbreaking rates across our treatments. The figure displays bar graphs for employee whistleblowing (left) and manager lawbreaking (right). To test for treatment effects, we follow our preregistration and estimate linear probability models with standard errors clustered at the session level. The results are presented in Table 1.

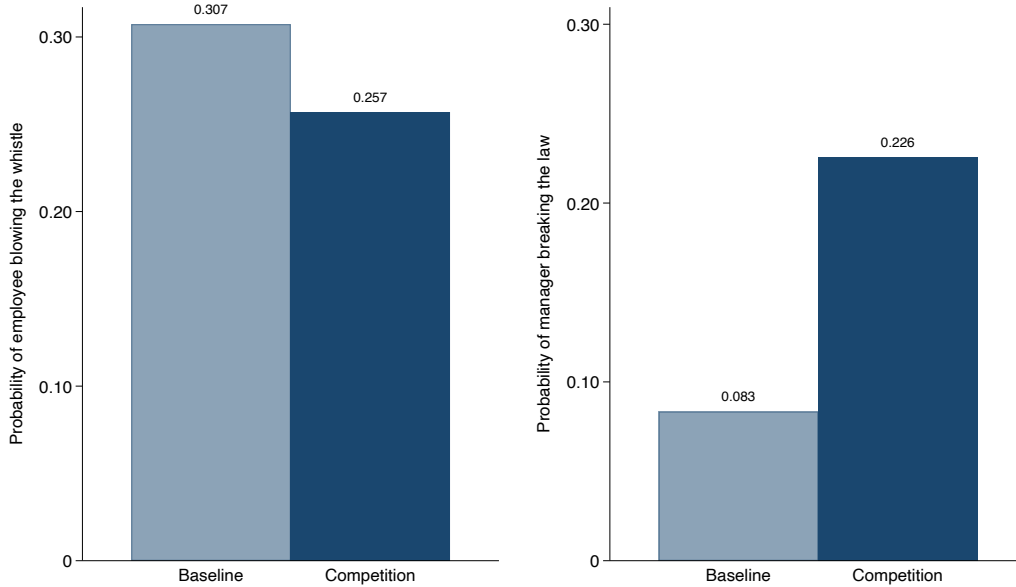


Figure 1: Whistleblowing (left) and lawbreaking (right) across treatments

Regarding employees' conditional willingness to report, 30.7% of employee decisions in the Baseline involved blowing the whistle if the manager had broken the law, compared to 25.7% in the Competition treatment.⁶ The econometric estimation, which controls for risk preferences, social value orientation, and demographics, reveals a small and non-significant decrease under competition ($\beta = -0.030, SE = 0.038, p = 0.432$, Table 1, Column 1). The point estimate corresponds to a 3 percentage point reduction, and the 95% confidence interval, $[-0.111, 0.049]$, ranges from a 11.1 percentage point decrease to a 4.9 percentage point increase. Thus, we find no evidence that competition substantially reduced employees' conditional willingness to blow the whistle.

⁶In our Baseline, we observe a higher whistleblowing rate than in Butler et al. (2020), possibly because our experiment involved repeated rounds whereas the original was a one-shot game.

Result 1. *Competition leads to a non-significant decrease in employees' conditional willingness to blow the whistle.*

While our primary focus is whistleblowing, we briefly examine manager lawbreaking. In the Baseline, 8.3% of managers chose to break the law, compared to 22.6% under Competition. The econometric estimation yields a sizeable but statistically imprecise increase in lawbreaking under competition ($\beta = 0.146$, $SE = 0.070$, $p = 0.056$, Table 1, Column 4). The point estimate corresponds to a 14.6 percentage point increase, with a 95% confidence interval ranging from a 0.4 percentage point decrease to a 29.5 percentage point increase. Thus, the estimate is economically meaningful and directionally consistent with Prediction 1, but the statistical evidence remains suggestive rather than conclusive.⁷

Result 2. *Competition leads to a directional and suggestive increase in manager lawbreaking.*

	Blow the whistle			Break the law		
	(1)	(2)	(3)	(4)	(5)	(6)
Competition	-0.030 (0.038)	0.016 (0.024)	0.006 (0.064)	0.146 (0.070)	0.043 (0.035)	-0.087 (0.089)
<i>Beliefs</i>						
Whistleblowing		0.707*** (0.046)	0.703*** (0.051)		-0.170 (0.134)	-0.393** (0.129)
Lawbreaking		-0.306** (0.099)	-0.313* (0.139)		0.705** (0.222)	0.885** (0.293)
Competition*Whistleblowing			0.012 (0.110)			0.514 (0.244)
Competition*Lawbreaking			0.017 (0.194)			-0.056 (0.293)
<i>Social appropriateness</i>						
Employee staying silent		-0.238 (0.124)	-0.236 (0.128)			
Manager lawbreaking					-0.089 (0.155)	0.015 (0.176)
Public losing earnings		-0.140 (0.075)	-0.139 (0.075)		0.401 (0.198)	0.382 (0.192)
Loyalty to firm		-0.130* (0.051)	-0.130* (0.049)		-0.108 (0.109)	-0.028 (0.135)
Observations	1,152	1,152	1,152	576	576	576

Standard errors in parentheses, clustered on matching group level. Controls: Age, Gender, Study, Risk, Social Value Orientation. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 1: The effect of competition, beliefs, and judgements on whistleblowing and lawbreaking

Additionally, we examine how beliefs and social appropriateness judgements correlate with observed behaviour. Whistleblowing is strongly positively correlated with beliefs about peer re-

⁷To test for possible dynamic effects, we split the data into the first six and last six rounds and reestimate the main specifications separately. For whistleblowing, the Competition coefficient is insignificant in both halves of the experiment ($p = 0.138$ for rounds 1–6 and $p = 0.900$ for rounds 7–12). For manager lawbreaking, the corresponding estimates remain suggestive but imprecise ($p = 0.053$ and $p = 0.097$). These results provide little indication that the main findings are driven by accumulating feedback or norm formation across rounds.

porting and negatively correlated with the expected frequency of lawbreaking (Table 1, Column 2). Similarly, managers are significantly more likely to break the law when they perceive such misconduct as prevalent among their peers (Table 1, Column 5). We find little evidence that social appropriateness judgements correlate with behaviour; employees blow the whistle less when they feel more loyal to their firm, while most other judgement measures are not robustly associated with choices.

Motivated by the strategy-method elicitation, we also estimate specifications interacting the Competition treatment directly with beliefs about whistleblowing and lawbreaking. These interactions test whether competition differentially affects decisions among participants with different beliefs about others’ behaviour. For employee whistleblowing, both interaction terms are small and statistically insignificant (Table 1, Column 3), suggesting that the null treatment effect is not concentrated among employees with particular beliefs about peer reporting or managerial lawbreaking. We therefore find that behaviour is strongly associated with beliefs in levels, but little evidence that beliefs moderate the effect of competition on employees’ conditional willingness to report.⁸

Observation 1. *Behaviour is strongly associated with beliefs about the prevalence of whistleblowing and lawbreaking, but only weakly associated with social appropriateness judgements. These beliefs do not appear to moderate the effect of competition on conditional whistleblowing.*

We further investigate whether the competition treatment directly altered beliefs and social appropriateness judgements. All participants provided the same belief and social appropriateness judgements. Table 2 reports treatment differences for the full sample and separately for managers and non-managers. Beliefs move descriptively in the direction of the observed behaviour, with lower expected whistleblowing and higher expected lawbreaking, although these differences do not survive the multiple hypothesis testing adjustment. In contrast, social appropriateness judgements remain relatively stable across treatments.

	All participants			Managers			Non-managers		
	Baseline	Competition	<i>p</i>	Baseline	Competition	<i>p</i>	Baseline	Competition	<i>p</i>
<i>Beliefs about frequency</i>									
Whistleblowing	0.335	0.277	0.545	0.258	0.262	1.000	0.354	0.280	0.570
Lawbreaking	0.314	0.328	1.000	0.197	0.315	0.388	0.343	0.331	1.000
<i>Social appropriateness</i>									
Employee staying silent	0.467	0.467	1.000	0.458	0.458	1.000	0.469	0.469	1.000
Manager breaking the law	0.281	0.290	1.000	0.312	0.240	0.591	0.273	0.302	1.000
Public losing earnings	0.165	0.204	0.453	0.167	0.219	1.000	0.164	0.201	0.557
Loyalty to firm	0.590	0.617	1.000	0.740	0.771	1.000	0.552	0.578	1.000
Observations	120	120	240	24	24	48	96	96	192

All entries are treatment means, standardised to lie between 0 and 1. Non-managers are participants who alternated between employee and member of the public roles. P-values are from OLS regressions of each outcome on the Competition treatment indicator, with standard errors clustered at the session level, and are adjusted using the Holm-Bonferroni procedure across the six belief and social appropriateness judgement outcomes within the corresponding participant group.

Table 2: Beliefs and social appropriateness judgements across treatments

⁸Each employee accounted for 6 out of 72 whistleblowing decisions in their session. Similarly, each manager accounted for 12 out of 36 lawbreaking decisions. Thus, stated belief may be biased towards participants’ own behaviour. Repeating the analysis using modified beliefs, i.e., the originally stated beliefs after subtracting each participant’s own decisions, provides qualitatively similar results.

Observation 2. *Beliefs move descriptively in the direction of observed behaviour, but these differences are not significant after adjustment for multiple testing; social appropriateness judgements remain relatively stable across treatments.*

3.2 Exploratory heterogeneity analysis

While our primary analysis shows limited aggregate treatment effects, we explore whether treatment effects differ across the Birmingham and Cambridge participant pools. To directly test for pool-level heterogeneity, we estimate the same linear probability models as in the preregistered analysis, adding an indicator for Cambridge and its interaction with the Competition treatment. These analyses were not preregistered and should therefore be interpreted cautiously.

For employees' conditional willingness to blow the whistle, the interaction between Competition and Cambridge is positive and statistically significant ($\beta = 0.137, SE = 0.056, p = 0.028$). This indicates that the negative effect of competition on whistleblowing is concentrated in the Birmingham pool: the estimated effect of Competition is negative for Birmingham participants ($\beta = -0.097, SE = 0.040, p = 0.029$), while the corresponding effect for Cambridge participants is close to zero. For manager lawbreaking, competition increases lawbreaking in the Birmingham pool ($\beta = 0.180, SE = 0.056, p = 0.006$), but the interaction between Competition and Cambridge is not statistically significant ($\beta = -0.069, SE = 0.118, p = 0.565$).

Thus, although the descriptive treatment effects differ across pools, we find clear statistical evidence of pool-level heterogeneity only for whistleblowing, not for manager lawbreaking. These exploratory results suggest that the aggregate null effect on whistleblowing masks heterogeneity across participant pools, but they should be viewed as suggestive boundary condition evidence rather than as a preregistered test of population differences.

Observation 3. *The exploratory participant pool analysis provides evidence of heterogeneous treatment effects on conditional whistleblowing, but not on manager lawbreaking.*

4 Concluding discussion

Across two participant pools, we find that employees' conditional willingness to blow the whistle is not systematically suppressed by competition. This null effect is informative given the structure of the design. Under the strategy-method elicitation, competition could reduce reporting through a simple material channel: if lawbreaking is expected to be more frequent, a conditional decision to report is more likely to become costly. Yet employees' own reporting commitments did not significantly decrease. Participants' beliefs moved in the predicted direction, but their actions and social appropriateness judgements were comparatively stable. This suggests that expectations about others' behaviour were more responsive to competition than employees' own conditional willingness to report misconduct.

If competition had systematically suppressed whistleblowing, it would suggest that interventions should be prioritised in highly competitive sectors where internal monitoring is most fragile. Our findings suggest that, in a punishment-based environment where reporting is elicited as a payoff-relevant conditional commitment, competition intensity does not by itself appear to strongly suppress willingness to report. This cautions against assuming that competitive

pressure necessarily undermines internal reporting, and consequently should be prioritised for intervention. This aligns with recent theoretical work, such as Thanassoulis (2023), which suggests that more competitive markets are not inherently more prone to misconduct. We emphasise that direct-response settings with observed misconduct remain an important avenue for future research.

The importance of these results is underscored by the statistical power of our design. With 240 participants and 80% power to detect a 10 percentage point reduction in whistleblowing, our findings offer a null result that is often missing due to publication bias. Moreover, our study includes two distinct participant pools, which allows us to explore whether the aggregate treatment effect masks pool-level heterogeneity. These exploratory analyses suggest that the aggregate null effect on whistleblowing may conceal meaningful differences across participant pools, although the evidence is not sufficient to establish the source or generality of this heterogeneity.

Documenting such heterogeneity is important for the scientific process and ties into the academic discussion of generalisability. A single-pool replication of our design could plausibly yield a different impression of the effect of competition depending on the participant pool studied, but larger replications across multiple pools would be needed to obtain more precise estimates of both the aggregate effect and the heterogeneity itself. Thus, our results should not be interpreted as showing that the corrosive influence of competition is necessarily population dependent. Rather, they suggest that the effect of competition on conditional whistleblowing may be context sensitive, and that apparent aggregate null effects can coexist with suggestive heterogeneity across samples. Further research is required to establish the specific conditions under which such effects emerge, prioritising larger and more diverse samples to move beyond the limitations of observations from a single participant pool.

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Appendix A Formal derivations of experimental predictions

We denote the probability with which the manager breaks the law in treatment j by b^j , and the probability with which the selected employee blows the whistle in treatment j by w^j , where $j \in \{B, C\}$ indicates the baseline and competition treatments respectively. We also denote by d_b^j and d_s^j respectively, the moral cost of the manager for breaking the law and the moral cost of the employee for not blowing the whistle in treatment j . Finally, we denote the expected firm surplus produced by the two employees before the manager makes any decision by S , and the probability that the manager can correctly solve the six multiplication problems by a .

A.1 Manager utility and lawbreaking comparative statics

In the Baseline treatment, breaking the law yields the manager their fixed income and half of the doubled surplus (S), while reducing utility by the moral cost of the act and the expected punishment.

$$U_b^B = 120 + S - 70w^B - d_b^B$$

Choosing the multiplication task yields the fixed income and half of the expected surplus, which is doubled only if the task is successfully completed.

$$U_t^B = 120 + \frac{1}{2} [a \times 2S + (1 - a) \times S] = 120 + \frac{1 + a}{2} S$$

The manager weakly prefers to break the law if $U_b^B \geq U_t^B$, which implies:

$$\frac{1 - a}{2} S \geq 70w^B + d_b^B \quad (1)$$

Intuitively, a manager is more likely to break the law if they are of low ability, if whistleblowing is less likely, or if their moral costs of doing so are low.

To derive the condition for the Competition treatment, we let π_{1S} denote the probability of winning the competition if the firm surplus is not doubled, and π_{2S} if it is doubled. By definition, $\pi_{2S} \geq \pi_{1S}$. Given that ties are broken randomly and initial surplus is expected to be uniform across firms⁹, π_{2S} is bounded below by $1/3$ (the case where all firms double their surplus) and π_{1S} is bounded above by $1/3$ (the case where no firms double their surplus). Thus, $\pi_{2S} \geq \frac{1}{3} \geq \pi_{1S}$.

In the Competition treatment, the utility from breaking the law is (after rearranging terms):

$$120 + \frac{3(1 + \pi_{2S})}{4} S - 70w^C - d_b^C$$

⁹The addition task is straightforward and almost all employees are expected to produce the maximal surplus. Out of 2,016 sets of the six addition problems that the employees in our experiment worked on, they solved all six correctly in 1,888 of them (93.6%) with almost all other instances resulting in five correct. Thus, the surplus of each firm before the manager decision was roughly fixed at 120, 60 from each employee.

The utility from the multiplication task is:

$$120 + \left[\frac{3a(1 + \pi_{2S})}{4} + \frac{3(1 - a)(1 + \pi_{1S})}{8} \right] S$$

The manager prefers to break the law if $U_b^C \geq U_t^C$, which simplifies to:

$$\frac{1 - a}{2} \left[\frac{3}{2} \left(\frac{1}{2} + \pi_{2S} - \frac{\pi_{1S}}{2} \right) \right] S \geq 70w^C + d_b^C \quad (2)$$

Since $\pi_{2S} \geq \frac{1}{3} \geq \pi_{1S}$, the term in the square brackets on the left-hand side is weakly larger than 1. This indicates that the expected material benefit of lawbreaking is higher under Competition than in the Baseline. Assuming $w^C \leq w^B$ and $d_b^C \leq d_b^B$, managers are predicted to break the law more frequently under competition.

A.2 Employee utility and whistleblowing comparative statics

For an employee, the decision is a contingent commitment: if the manager broke the law, would the employee blow the whistle? The expected monetary cost of committing to report is $0.5 \times 25 \times b^j$, because the reporting cost is incurred only if the manager chose to break the law and the employee's decision is randomly selected for implementation. The parameter d_s^j captures the moral cost of committing not to report in the state in which the manager broke the law. The following comparison should therefore be interpreted as a benchmark for the implemented strategy-method elicitation, not as a full psychological model of reporting after directly observed misconduct.

In the Baseline, the employee prefers to report if:

$$d_s^B \geq \frac{25}{2} b^B \quad (3)$$

In the Competition treatment, the condition is:

$$d_s^C \geq \frac{25}{2} b^C \quad (4)$$

Given Prediction 1 ($b^C \geq b^B$), the expected cost of reporting is higher under competition. Consequently, we predict that employees will be less likely to blow the whistle in the Competition treatment, a tendency that would be further amplified if competition also reduces the moral cost of silence ($d_s^C \leq d_s^B$).

Appendix B Additional results

This appendix section reports descriptive cell means for observable characteristics and task performance across treatment–participant–pool cells. We do not report formal balance test p -values, since treatment assignment was randomised and multiple individual tests are difficult to interpret. Multiplication task performance is descriptive only, as it is observed conditional on managers choosing the task.

Treatment	Location	N	Age	Female	Social	Natural	Risk	SVO
Baseline	Birmingham	60	22.40	0.57	0.22	0.20	1.60	21.75
Baseline	Cambridge	60	22.40	0.50	0.28	0.23	1.95	22.33
Competition	Birmingham	60	23.07	0.52	0.23	0.22	1.55	20.32
Competition	Cambridge	60	22.50	0.63	0.33	0.23	2.12	23.36

Each row corresponds to one treatment–participant–pool cell. Entries are cell means, except for N . Female, Social., and Natural are proportions. Social. includes participants in economics and social sciences, while Natural denotes participants in natural sciences. The omitted gender categories are male and other/prefer not to say. The omitted field-of-study categories are humanities, applied sciences, and other. Risk denotes the chosen lottery in the risk-preference task, with higher values corresponding to greater risk taking. SVO denotes the participant’s Social Value Orientation score.

Table B1: Observable characteristics across treatments and participant pools

Treatment	Location	Task	N	Correct	SD
Baseline	Birmingham	Addition	576	5.94	0.32
Baseline	Cambridge	Addition	576	5.90	0.50
Competition	Birmingham	Addition	576	5.95	0.23
Competition	Cambridge	Addition	576	5.91	0.36
Baseline	Birmingham	Multiplication	143	4.78	1.24
Baseline	Cambridge	Multiplication	121	3.79	1.72
Competition	Birmingham	Multiplication	117	4.94	1.46
Competition	Cambridge	Multiplication	106	4.54	1.35

Each row corresponds to one treatment–participant–pool–task cell. Correct denotes the mean number of correctly solved problems out of six, and SD denotes the corresponding standard deviation. Addition-task observations pool rounds in which participants were assigned the role of employee or member of the public. Multiplication-task observations include only managers who chose to perform the multiplication task rather than break the law. These observations are therefore conditional on the manager’s decision and are not used as an ex ante balance measure. Consequently, N differs across tasks and treatment–participant–pool cells.

Table B2: Task performance across treatments and participant pools

Appendix C Instructions

C.1 Welcome

Welcome to this experiment. Please read the following instructions carefully. We ask that you do not communicate with other participants during the experiment. The use of mobile phones is not allowed during this experiment. If you have any questions, or need assistance of any kind, at any time, an experimenter will assist you privately. The data collected through this experiment does not include your name or any other information that would allow your identification. All the data you provide during the experiment cannot be traced back to you.

In addition to your participation fee, you may earn substantially more money from today's experiment. You will be paid privately and anonymously in cash at the end of the experimental session today. Earnings during the experiment will be denominated in Experimental Currency Units, or ECU. Each ECU is worth £0.01. The participation fee is £2.00. After the experiment finishes, you will be paid the money you earned plus your participation fee.

Today's experiment consists of 12 rounds. In all rounds, you will play the same game, but your role may change in every round. Instructions for the game and the roles will be shown to you on your computer screen.

C.2 General instructions

The experiment consists of 12 rounds. In each round, there will be 3 firms, each consisting of 1 manager and 2 employees, and 6 members of the public.

In the beginning of the experiment, the 3 managers of the 3 firms will randomly be drawn. The managers will keep their role for the whole experiment.

In each round, the remaining participants will randomly be assigned either the role of employee or of member of the public. The employees will randomly be assigned to a firm. The matching ensures that employees and members of the public will swap roles in every other round.

All the instructions and rules are also printed in the page you have in front of you.

C.3 Rules of the game

Employee Task: The two employees have the task of adding two numbers and report their sum. Each correct answer gives them 20 points. Additionally each correct answer by each employee generates 10 points to the firm surplus. In total, each employee will be given 6 pairs of numbers to add. Employees have 120 seconds to solve as many as they can.

Manager Decision & Task: Managers receive a fixed income of 120 points. Managers have the opportunity to double the firm surplus. Their decision is to choose how they want to try to double the surplus. They can do so in two ways:

- Do the Manager Task: In the task, managers are asked to multiply two numbers and report their product. Managers will be given 6 pairs of numbers to multiply. Managers

have 120 seconds to solve as many as they can. If they give at least 3 correct answers, the firm surplus is doubled.

- Break the law: If the manager breaks the law, they skip the Manager Task and the firm surplus will be doubled automatically. However, it will also generate a loss of 20 points to each of the 6 members of the public.

Winning firm: One of the three firms will win the **competition** and their surplus will be further **increased by 50%**. Each of the other two firms that did not win the competition will have their surplus **reduced by 25%**. The winning firm will be the firm with the **largest surplus**. [COMPETITION TREATMENT ONLY]

Surplus: The total firm surplus will be distributed as follows: The manager keeps 50% of the surplus, and each employee will keep 25%.

Employee decision: The employees have the option to blow the whistle if their manager broke the law. The decision to blow the whistle will be relevant **ONLY** if the manager broke the law. If the manager did not break the law, this decision will not be implemented. Employees make this decision after the manager's decision has been made, but without observing whether the manager chose the multiplication task or chose to break the law. Each employee can decide whether they are willing to blow the whistle. Blowing the whistle will cost the selected employee 25 points, and will generate a penalty of 70 points to the manager. This penalty will be removed from the manager's earnings. One of the two employees will randomly be selected and their decision will be implemented. The decision of the other employee will not be implemented.

Members of the Public Task: The members of the public have the task of adding two numbers and report their sum. Each correct answer gives them 20 points. In total, each member of the public will be given 6 pairs of numbers to add. Members of the public have 120 seconds to solve as many as they can.

Payoff summary

- Employees:
 - Earn 20 points for each correct answer in number-adding task.
 - Earn 25% of the final firm surplus.
 - Lose 25 points if they blow the whistle and their decision is implemented.
- Managers:
 - Earn fixed income of 120 points.
 - Earn 50% of the final firm surplus.
 - Lose 70 points if they broke the law and the selected employee blew the whistle.
- Members of the public:

- Earn 20 points for each correct answer in number-adding task.
- Lose 20 points for each manager that broke the law.

Paid rounds: At the end of the session, 4 rounds will be randomly selected for payment.

C.4 Numerical example

C.4.1 Baseline treatment

- **Firm A:** Each employee answered 5 questions correctly. Since each correct answer generates 10 points to the firm surplus, the initial surplus equals 100. The manager broke the law and therefore the surplus was automatically doubled to 200. The selected employee chose to blow the whistle.
 - **Selected employee:** earns 125 points (+100 points from 5 correct answers, +50 points from 25% of the firm surplus, –25 points from blowing the whistle).
 - **Other employee:** earns 150 points (+100 points from 5 correct answers, +50 points from 25% of the firm surplus). The second employee was not selected, so their decision to blow the whistle was not implemented and did not cost them points.
 - **Manager:** earns 150 points (+120 points as fixed income, +100 points from 50% of the firm surplus, –70 points because the selected employee blew the whistle).
- **Firm B:** The first employee answered 5 questions correctly. The second employee answered 4 questions correctly. The initial surplus equals 90. The manager did not break the law and engaged in the multiplication task. The manager gave 4 correct answers, so the surplus was doubled to 180. The first employee was selected and chose to blow the whistle.
 - **Selected employee:** earns 145 points (+100 points from 5 correct answers, +45 points from 25% of the firm surplus). The manager did not break the law, so the decision to blow the whistle was not implemented.
 - **Other employee:** earns 125 points (+80 points from 4 correct answers, +45 points from 25% of the firm surplus). The second employee was not selected, so their decision to blow the whistle was not implemented and did not cost them points.
 - **Manager:** earns 210 points (+120 points as fixed income, +90 points from 50% of the firm surplus). The manager did not break the law, so they did not lose any points.
- **Firm C:** The first employee answered 6 questions correctly. The second employee answered 6 questions correctly. The initial surplus equals 120. The manager did not break the law and engaged in the multiplication task. The manager gave 2 correct answers, so the surplus was not doubled. The first employee was selected and chose to blow the whistle.

- **Selected employee:** earns 150 points (+120 points from 6 correct answers, +30 points from 25% of the firm surplus). The manager did not break the law, so the decision to blow the whistle was not implemented.
 - **Other employee:** earns 150 points (+120 points from 6 correct answers, +30 points from 25% of the firm surplus). The second employee was not selected, so their decision to blow the whistle was not implemented and did not cost them points.
 - **Manager:** earns 180 points (+120 points as fixed income, +60 points from 50% of the firm surplus). The manager did not break the law, so they did not lose any points.
- **Members of the public:** One manager (the manager in Firm A) broke the law. Therefore, each member of the public loses 20 points. In addition, each member earns 20 points for each correct answer in their own number-adding task.

C.4.2 Competition treatment

- **Firm A:** Each employee answered 5 questions correctly. The initial surplus equals 100. The manager broke the law and therefore the surplus was automatically doubled to 200. The other firms have surpluses of 180 and 120, so Firm A won the competition and its surplus was increased by 50% to 300. The selected employee chose to blow the whistle.
 - **Selected employee:** earns 150 points (+100 points from 5 correct answers, +75 points from 25% of the final firm surplus, –25 points from blowing the whistle).
 - **Other employee:** earns 175 points (+100 points from 5 correct answers, +75 points from 25% of the final firm surplus). The second employee was not selected, so their decision to blow the whistle was not implemented and did not cost them points.
 - **Manager:** earns 200 points (+120 points as fixed income, +150 points from 50% of the final firm surplus, –70 points because the selected employee blew the whistle).
- **Firm B:** The first employee answered 5 questions correctly. The second employee answered 4 questions correctly. The initial surplus equals 90. The manager did not break the law and engaged in the multiplication task. The manager gave 4 correct answers, so the surplus was doubled to 180. The other firms have surpluses of 200 and 120, so Firm B did not win the competition and its surplus was reduced by 25% to 135. The first employee was selected and chose to blow the whistle.
 - **Selected employee:** earns 133.75 points (+100 points from 5 correct answers, +33.75 points from 25% of the final firm surplus). The manager did not break the law, so the decision to blow the whistle was not implemented.
 - **Other employee:** earns 113.75 points (+80 points from 4 correct answers, +33.75 points from 25% of the final firm surplus). The second employee was not selected, so their decision to blow the whistle was not implemented and did not cost them points.
 - **Manager:** earns 187.5 points (+120 points as fixed income, +67.5 points from 50% of the final firm surplus). The manager did not break the law, so they did not lose any points.

- **Firm C:** The first employee answered 6 questions correctly. The second employee answered 6 questions correctly. The initial surplus equals 120. The manager did not break the law and engaged in the multiplication task. The manager gave 2 correct answers, so the surplus was not doubled. The other firms have surpluses of 200 and 180, so Firm C did not win the competition and its surplus was reduced by 25% to 90. The first employee was selected and chose to blow the whistle.
 - **Selected employee:** earns 142.5 points (+120 points from 6 correct answers, +22.5 points from 25% of the final firm surplus). The manager did not break the law, so the decision to blow the whistle was not implemented.
 - **Other employee:** earns 142.5 points (+120 points from 6 correct answers, +22.5 points from 25% of the final firm surplus). The second employee was not selected, so their decision to blow the whistle was not implemented and did not cost them points.
 - **Manager:** earns 165 points (+120 points as fixed income, +45 points from 50% of the final firm surplus). The manager did not break the law, so they did not lose any points.
- **Members of the public:**
 - One manager (the manager in Firm A) broke the law. Therefore, each member of the public loses 20 points. In addition, each member earns 20 points for each correct answer in their own number-adding task.

Understanding questions

1. **How many rounds will be paid?**
 - ☐ All rounds
 - ☐ Four randomly selected rounds
2. **The manager broke the law. What happens?**
 - ☐ Firm surplus gets doubled, members of the public do not lose points
 - ☐ Firm surplus gets doubled, members of the public lose points
 - ☐ Firm surplus does not get doubled, members of the public do not lose points
 - ☐ Firm surplus does not get doubled, members of the public lose points
3. **The manager did not break the law. They solved 5 multiplication questions correctly. What happens?**
 - ☐ Firm surplus gets doubled, members of the public do not lose points
 - ☐ Firm surplus gets doubled, members of the public lose points
 - ☐ Firm surplus does not get doubled, members of the public do not lose points
 - ☐ Firm surplus does not get doubled, members of the public lose points

4. **The manager did not break the law. They solved 6 multiplication questions correctly. What happens?**

- ☐ Firm surplus gets doubled, members of the public do not lose points
- ☐ Firm surplus gets doubled, members of the public lose points
- ☐ Firm surplus does not get doubled, members of the public do not lose points
- ☐ Firm surplus does not get doubled, members of the public lose points

5. **The manager broke the law. Employee A was selected and blew the whistle. Employee B was not selected and did not blow the whistle. What happens?**

- ☐ Employee A loses points, Employee B loses points, manager loses points
- ☐ Employee A does not lose points, Employee B loses points, manager loses points
- ☐ Employee A loses points, Employee B does not lose points, manager loses points
- ☐ Employee A loses points, Employee B loses points, manager does not lose points
- ☐ Employee A does not lose points, Employee B does not lose points, manager loses points
- ☐ Employee A does not lose points, Employee B loses points, manager does not lose points
- ☐ Employee A loses points, Employee B does not lose points, manager does not lose points
- ☐ Employee A does not lose points, Employee B does not lose points, manager does not lose points

6. **The manager did not break the law. Employee A was selected and blew the whistle. Employee B was not selected and did not blow the whistle. What happens?**

- ☐ Employee A loses points, Employee B loses points, manager loses points
- ☐ Employee A does not lose points, Employee B loses points, manager loses points
- ☐ Employee A loses points, Employee B does not lose points, manager loses points
- ☐ Employee A loses points, Employee B loses points, manager does not lose points
- ☐ Employee A does not lose points, Employee B does not lose points, manager loses points
- ☐ Employee A does not lose points, Employee B loses points, manager does not lose points
- ☐ Employee A loses points, Employee B does not lose points, manager does not lose points
- ☐ Employee A does not lose points, Employee B does not lose points, manager does not lose points

7. **In each round, one of the firms will win the competition. How will the winning firm be determined?** [COMPETITION TREATMENT ONLY]

- ☐ The winning firm is randomly determined
- ☐ The winning firm will be the firm with the largest surplus

C.5 Manager decision

You are the manager of a firm.

The other two members of your firm have been assigned the role of employees. Each employee will now engage in the number-addition task. Each correct answer gives the employee 20 points and generates 10 points to the firm surplus. The total firm surplus created by the employees will later be distributed among all members of the firm.

As a manager, you receive a fixed payment of 120 points. Additionally, you will receive payment from the distribution of the firm surplus. The total firm surplus created by the employees will be distributed as follows: The manager keeps 50% of the surplus. Each employee keeps 25% of the surplus.

As a manager, you have the opportunity to double the firm surplus your employees created. If the surplus gets doubled, payments from the surplus to both you and the employees will also double. There are two ways in which you can double the surplus:

- **Manager task:** You will be given 6 pairs of numbers and asked to report their product. If you answer at least 3 multiplication problems correctly, the firm surplus will be doubled.
- **Break the law:** If you decide to break the law, the firm surplus will be doubled automatically without completing the multiplication task. However, this will generate a loss of 20 points to each of the 6 members of the public.

There are two other firms in the session, each with one manager and two employees. They are now facing the same decisions as you are. Note that:

- Both employees will later decide whether they would blow the whistle if you decided to break the law. One of the two employees will be randomly selected, and only that employee's decision will be implemented.
- Blowing the whistle will cost the selected employee 25 points and will generate a penalty of 70 points to you. This penalty will be deducted from your earnings.

Please decide whether you would like to engage in the multiplication task or whether you would prefer to break the law.

- ☐ Multiplication task
- ☐ Break the law

C.6 Employee task

You are an employee of a firm. The other employee of the firm is facing the same decision task as you.

You and the other employee will now engage in a task that will generate earnings for yourself and the firm. The task of the employees is to add two numbers and report their sum. You will be given 6 pairs of numbers to add. Each correct answer gives you 20 points and generates 10 points to the firm surplus. The surplus of the firm is the total number of correct answers given by both employees multiplied by 10.

The manager of the firm will receive a fixed income of 120 points and will engage in a different task. The task of the manager is to multiply two numbers and report their product. If the manager solves at least 3 out of 6 multiplication problems correctly, the firm surplus will be doubled. The total firm surplus will be distributed as follows: The manager keeps 50% of the surplus. Each employee keeps 25% of the surplus.

C.7 Manager task [IF MANAGER CHOSE TASK]

You are the manager of a firm.

The other two members of your firm have been assigned the role of employees. Each employee will now engage in the number-addition task. Each correct answer gives the employee 20 points and generates 10 points to the firm surplus. The total firm surplus created by the employees will later be distributed among all members of the firm.

You as the manager of the firm will receive a fixed wage of 120 points and will engage in a different task. Your task will be to multiply two numbers together and report their product. If you solve at least 3 out of 6 problems correctly, the surplus of the firm will be doubled. The total firm surplus will be distributed as follows: The manager will keep 50% of the surplus, and each employee will keep 25%.

C.8 Employee decision

You are an employee of a firm. You and the other employee generated a firm surplus.

While you were engaged in the number-addition task, your manager was given the chance to double the firm surplus, by engaging in a number-multiplication task and solve at least 3 out of 6 problems correctly. Alternatively, the manager could skip the multiplication task and double the surplus by breaking the law. If the manager chose to break the law, they knew that the surplus would be automatically doubled, and that their decision to break the law would reduce the earnings of each member of the public by 20 points. All employees and managers faced an identical situation as you do now.

You will now be given to the chance to blow the whistle on your manager if they broke the law. Either you or the other employee will randomly be selected and, if your manager broke the law, the whistleblowing decision below will be implemented. Note that:

- Blowing the whistle will cost you 25 points, and will generate a penalty of 70 points to the manager. This penalty will be removed from the manager's earnings.

If the manager broke the law, what would you like to do?

- ☐ I would like to blow the whistle.
- ☐ I would not like to blow the whistle.

C.9 Feedback

- **Role**

- In this round, your role was: *Employee*. [EMPLOYEES ONLY]
- In this round, your role was: *Manager*. [MANAGERS ONLY]
- In this round, your role was: *Member of the Public*. [PUBLIC ONLY]

- **Performance**

- You gave X correct answers. [EMPLOYEES ONLY]
- You gave X correct answers. [MANAGERS ONLY, IF MANAGER CHOSE TASK]
- You gave X correct answers. [PUBLIC ONLY]

- **Firm Ranking**

- Your firm had the largest surplus and won the competition. If this round is selected for payment, your firm surplus will be increased by 50%. [COMPETITION TREATMENT, WINNING FIRM MEMBERS ONLY]
- Your firm did not have the largest surplus and did not win the competition. If this round is selected for payment, your firm surplus will be decreased by 25%. [COMPETITION TREATMENT, NON-WINNING FIRM MEMBERS ONLY]

C.10 Survey

Estimates

Please answer the following questions. For each question, you will receive additional 20 points if your answer is within a range of 3 of the actual number. Your earnings from this task will be added to your earnings from the game.

- The managers had the choice between breaking the law and doing the multiplication task 36 times in total during the experiment. How many times do you think they broke the law?
- The employees had the choice to blow the whistle or not 72 times in total during the experiment. How many times do you think they blew the whistle?

Social appropriateness judgements

Please answer the following questions. For each question, you will receive additional 10 points if your answer matches the answer of the majority of the participants in today's experiment. Your earnings from this task will be added to your earnings from the game.

- How would you judge a manager who broke the law?
(Very immoral, Immoral, No judgement, Moral, Very moral)
- How would you judge an employee who did not blow the whistle?
(Very immoral, Immoral, No judgement, Moral, Very moral)
- How do you feel about the members of the public losing earnings because managers broke the law?
(Very unacceptable, Unacceptable, No judgement, Acceptable, Very acceptable)
- In the rounds that you were part of a firm, how loyal did you feel to the firm?
(Not loyal at all, Not very loyal, Neutral, Loyal, Very loyal)

Lottery task

In the following task, 5 different lotteries are presented on your screen. In each of these lotteries, both rewards A and B are equally likely, i.e. have a probability of exactly 50%. The rewards are denoted in points.

You are asked to choose exactly one of the lotteries, which subsequently will be implemented. A random generator will determine whether you win reward A or reward B, respectively. At the end of the experiment, your reward will be added to you earnings.

	Lottery 1	Lottery 2	Lottery 3	Lottery 4	Lottery 5
Reward A (50%)	140	120	100	80	60
Reward B (50%)	140	180	220	260	300

Distribution task

In this task you have been randomly paired with another person, whom we will refer to as the other. This other person is someone you do not know and will remain mutually anonymous. All of your choices are completely confidential. You will be making a series of 6 decisions about allocating resources between you and this other person. For each of the following questions, please indicate the distribution you prefer most by choosing the button along the midline. You can only choose one distribution for each of the 6 questions. Your decisions will yield money for both yourself and the other person.

There are no right or wrong answers, this is all about personal preferences. One of the 6 decisions will randomly be selected and implemented. At the end of the experiment, the outcome will be added to you earnings.

Option	1	2	3	4	5	6	7	8	9
You receive	85	85	85	85	85	85	85	85	85
Other receives	85	76	68	59	50	41	33	24	15

Option	1	2	3	4	5	6	7	8	9
You receive	85	87	89	91	93	94	96	98	100
Other receives	15	19	24	28	33	37	41	46	50

Option	1	2	3	4	5	6	7	8	9
You receive	50	54	58	63	68	72	76	81	85
Other receives	100	98	96	94	93	91	89	87	85

Option	1	2	3	4	5	6	7	8	9
You receive	50	54	59	63	68	72	76	81	85
Other receives	100	89	79	68	58	47	36	26	15

Option	1	2	3	4	5	6	7	8	9
You receive	100	94	88	81	75	69	63	56	50
Other receives	50	56	63	69	75	81	88	94	100

Option	1	2	3	4	5	6	7	8	9
You receive	100	98	96	94	93	91	89	87	85
Other receives	50	54	59	63	68	72	76	81	85

Demographics

Please enter the following information.

- Please indicate your age.
- Please indicate your field of study.
(Economics, Social Sciences, Natural Sciences, Humanities, Applied Sciences, Other)
- Please indicate your gender.
(Male, Female, Prefer not to answer)